

Create a honeydoc:

- Implant a web bug in a Microsoft Word document and deploy a minimal Apache web server to monitor communication from wherever someone opens the document

Configuring the Honeypot:

- Create a web server to listen for connections from opened documents by using a minimal Apache configuration.
- Since it's likely attackers will pull files off the network before opening them, this Apache server should be accessible from the public internet so the document HTTP request will reach the server for logging.
- Ideally, deploy the server away from your network infrastructure by using a cloud service

On an Ubuntu system:

1. Install the apache2 package:

sudo apt-get install apache2

2. If you've already explicitly enabled the system firewall, allow inbound access to port 80:

sudo ufw allow 'Apache'

3. Create a directory for your domain. I'll identify this server with the generic and benign name 'update':

sudo mkdir /var/www/update

4. Assign ownership of the directory to a non-root user:

sudo chown -R \$USER:\$USER /var/www/update

5. Create an index page in the web directory:

nano /var/www/update/index.html

6. You can put anything you like on this page, but keep in mind an attacker might eventually browse here. The web server serves this content when someone requests the webroot. A simple blank page is a good place to start:

- **<html>**
- **<head>**
- **<title>Update Server</title>**
- **</head>**
- **<body></body>**
- **</html>**

Create a new virtual host file:

sudo nano /etc/apache2/sites-available/update.conf

7. Insert the following block into the file, substituting appropriate values for the naming of your web server. Keep in mind that you'll use the URL in `ServerName` and `ServerAlias` to reach the server. `DocumentRoot` should point to the directory you created in step 3.

- **<VirtualHost *:80>**
- **ServerAdmin me@ihulDS.gr**
- **ServerName update.ihulDS.gr**
- **ServerAlias update.ihulDS.gr**
- **DocumentRoot /var/www/update**
- **ErrorLog \${APACHE_LOG_DIR}/error.log**
- **CustomLog \${APACHE_LOG_DIR}/access.log combined**
- **</VirtualHost>**
- Enable the virtual host:

sudo a2ensite update.conf

8. Disable the default website:

- **sudo a2dissite 000-default.conf**

9. Restart Apache:

sudo systemctl restart apache2

10. Verify the server is running by connecting to the `ServerName` URL in a browser

- You should see a blank page with the Update Server title created in step 6.

11. With the server running, you'll need a custom URL for the honeydoc we're about to create. That should consist of the web server URL plus a token. For example,

12. <http://update.ihulDS.gr/7cba9f2>

13. The string '7cba9f2' is a randomly generated token that doesn't have any meaning on its own. However, it does help you identify where an attacker is on your network because it's unique to a single document that ideally exists only in a single place

14. Browsing to <http://update.ihuids.org/7cba9f2> (or whatever your version of this token is) yields an HTTP 404 Not Found response from the web server because nothing exists at that URL

- With the web server listening and a token link in hand, we're ready to configure the honeydoc. A honeydoc starts as a typical Office document that we'll modify

to include a web bug. Follow these steps in Microsoft Word for Office 365 on a Windows 10 system

1. Create a new blank document or open an existing document.
2. Click the **Insert** tab, select **Footer**, and choose one of the footer options to add the footer to the document. Or, if your document already has a footer, double click in the footer area to edit it. You can add the web bug anywhere, but I prefer the header or footer because it's less likely to be accidentally removed by changes made in other parts of the document.
3. From the **Header & Footer** menu, select **Quick Parts** and **Field**.
4. Select **IncludePicture** from the **Field names** list. Configure the following options and click **OK**:
 - In the **Filename or URL** textbox, type the token link created after setting up the web server.
 - Enable the **Data not stored with document** and **Preserve formatting during updates** checkboxes.
5. An empty picture box should load into the footer area. Double click it to switch to the **Picture Format** toolbar. Change the **Height** and **Width** options to .01”.

This change makes the web bug virtually invisible to the eye⁵. You can still select it by clicking and dragging your mouse over the area where it's located.

6. Save and close the document.
7. Whenever you open the document the next time, Word attempts to load the image from the remote URL, generating a request to your web server.
8. Verify if the process worked by looking for an entry containing the token in **/var/log/apache2/access.log**

Delivery

Create a pdf file and name it IALect2_xxxxxxxxx where xxxxxxxxx is your name

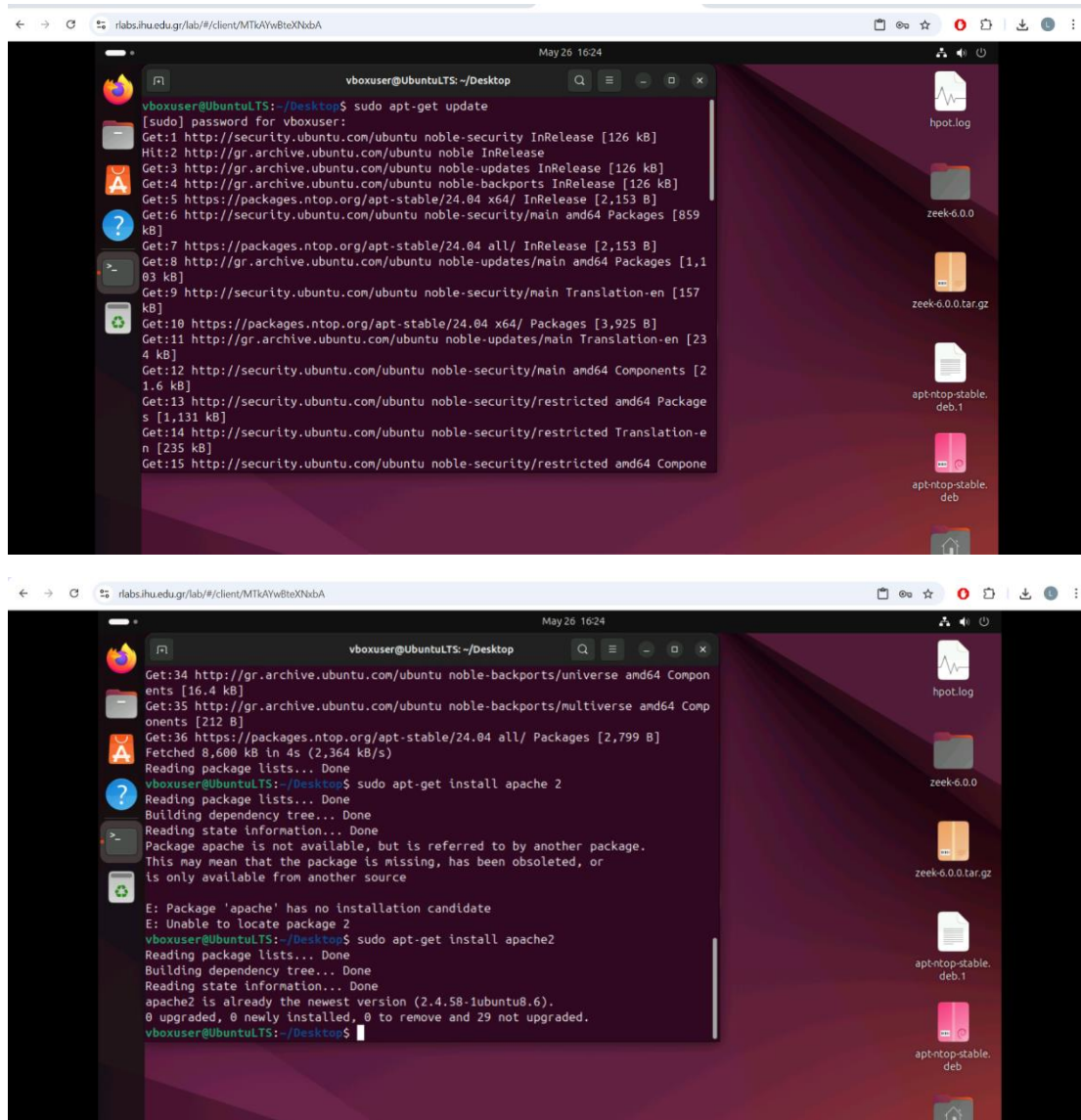
PS. Provide screenshots.

PS: Some steps I've already completed during the previous lecture that's why they exist already

Step 1: Install Apache

`sudo apt-get update`

`sudo apt-get install apache2`



```
vboxuser@UbuntuLTS: ~/Desktop
[sudo] password for vboxuser:
Get:1 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Hit:2 http://gr.archive.ubuntu.com/ubuntu noble InRelease
Get:3 http://gr.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://gr.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 https://packages.ntop.org/apt-stable/24.04 x64/ InRelease [2,153 B]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [859 kB]
Get:7 https://packages.ntop.org/apt-stable/24.04 all/ InRelease [2,153 B]
Get:8 http://gr.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,103 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [157 kB]
Get:10 https://packages.ntop.org/apt-stable/24.04 x64/ Packages [3,925 B]
Get:11 http://gr.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [234 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1,131 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [235 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [16.4 kB]
Get:16 http://gr.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:17 https://packages.ntop.org/apt-stable/24.04 all/ Packages [2,799 B]
Fetched 8,600 kB in 4s (2,364 kB/s)
Reading package lists... Done
vboxuser@UbuntuLTS: ~/Desktop$ sudo apt-get install apache 2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package apache is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
E: Package 'apache' has no installation candidate
vboxuser@UbuntuLTS: ~/Desktop$ sudo apt-get install apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
apache2 is already the newest version (2.4.58-1ubuntu8.6).
0 upgraded, 0 newly installed, 0 to remove and 29 not upgraded.
vboxuser@UbuntuLTS: ~/Desktop$
```

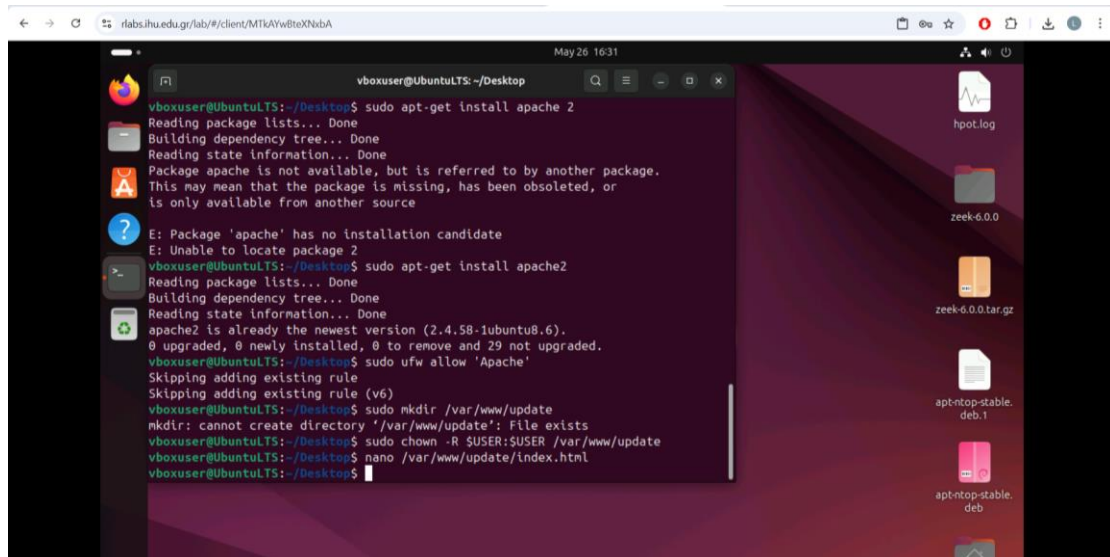
Step 2: Allow HTTP Through UFW (Firewall)

```
sudo ufw allow 'Apache'
```

Step 3: I created a Web Directory

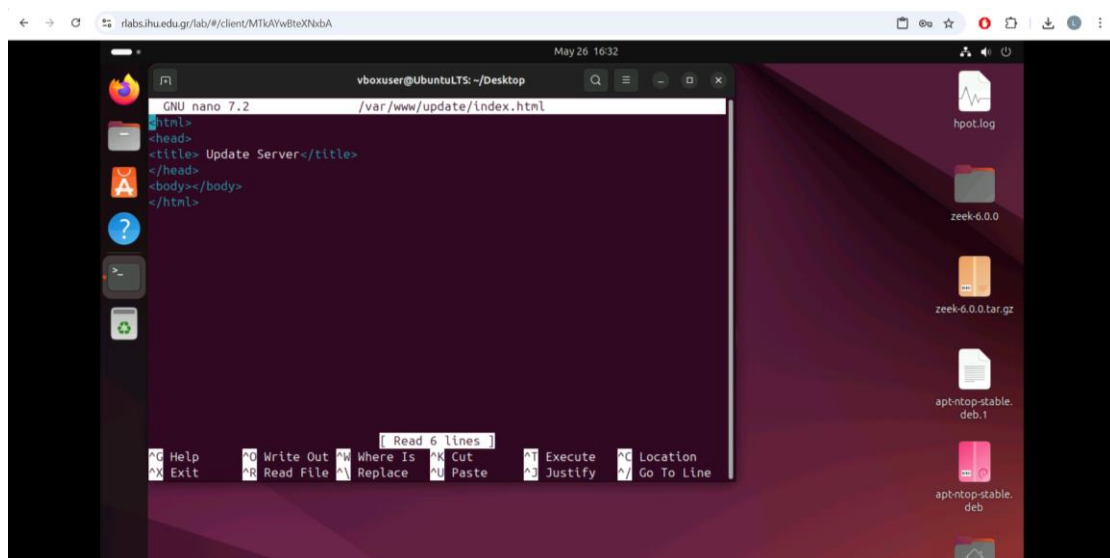
```
sudo mkdir /var/www/update
```

```
sudo chown -R $USER:$USER /var/www/update
```



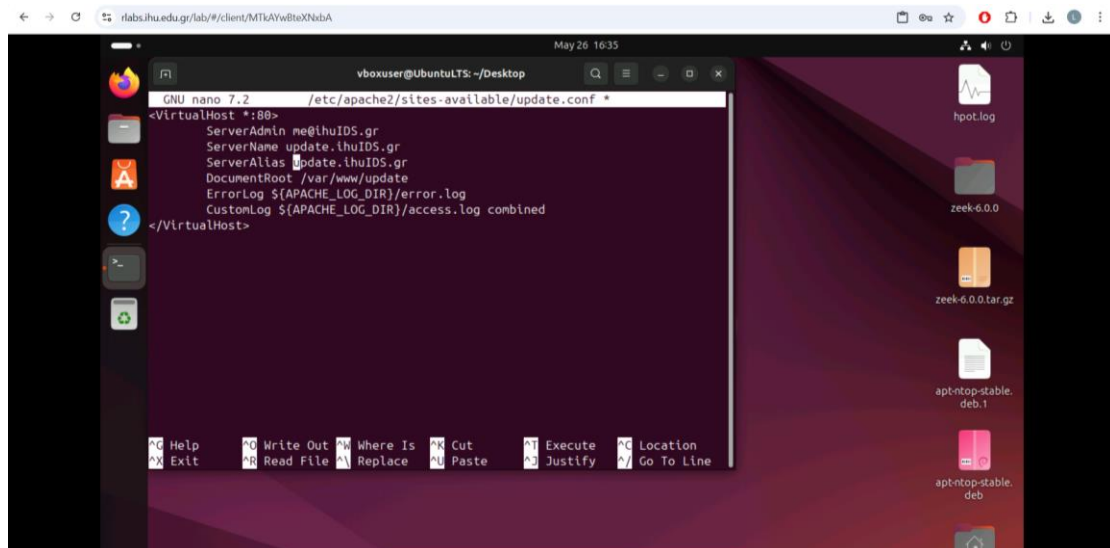
Step 4: I created the index.html File

```
nano /var/www/update/index.html
```



Step 5: I created an Apache Virtual Host Config

```
sudo nano /etc/apache2/sites-available/update.conf
```

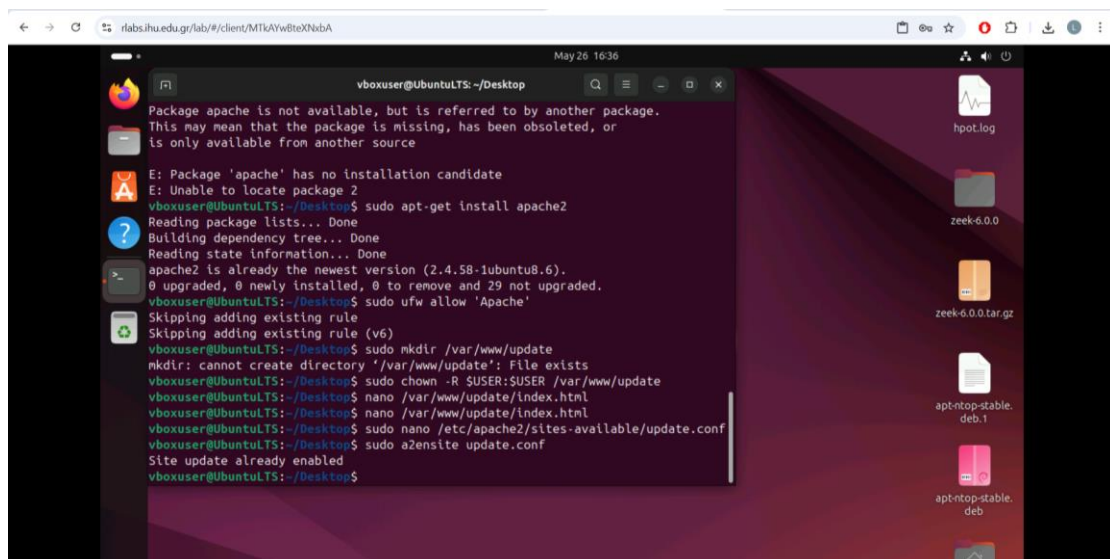


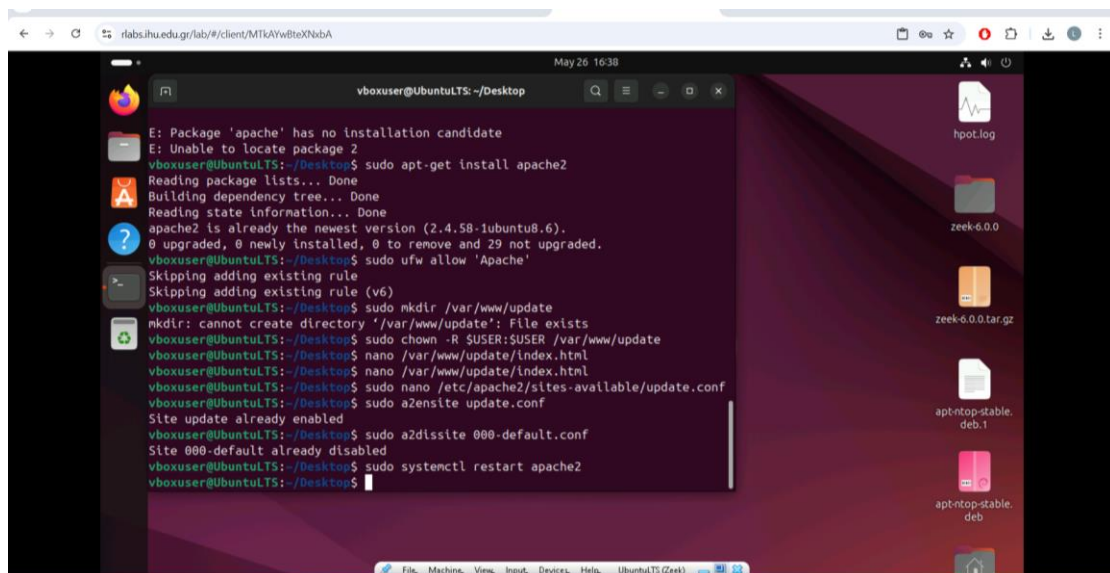
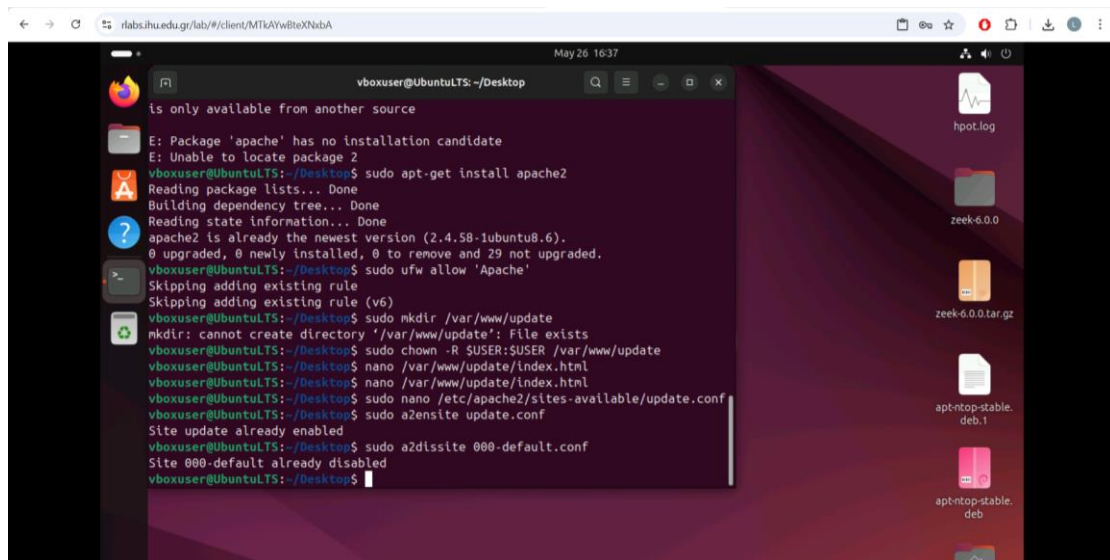
Step 6: Enable and Restart

sudo a2ensite update.conf

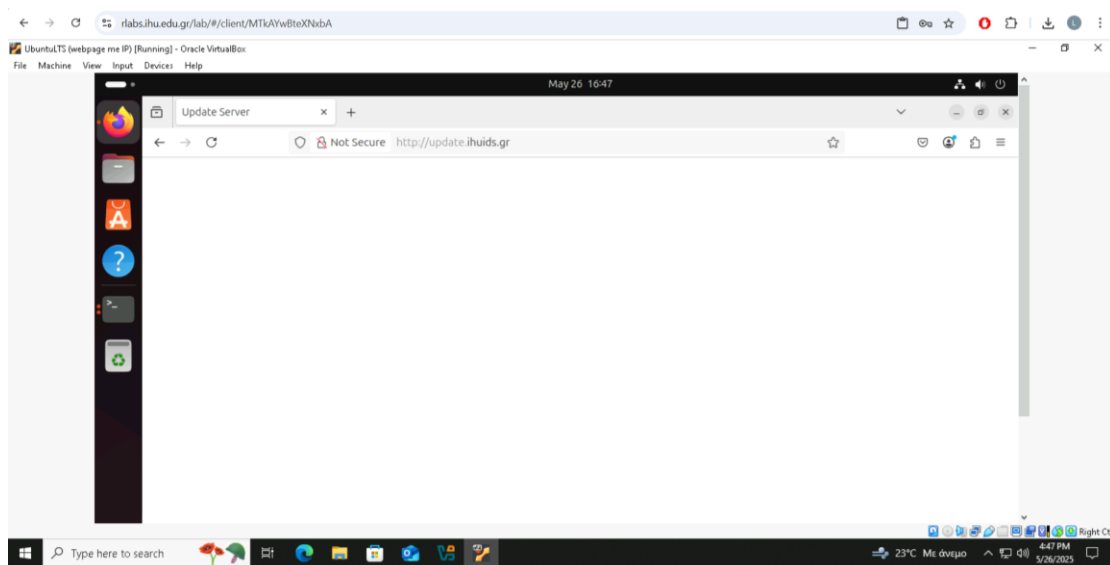
sudo a2dissite 000-default.conf

sudo systemctl restart apache2



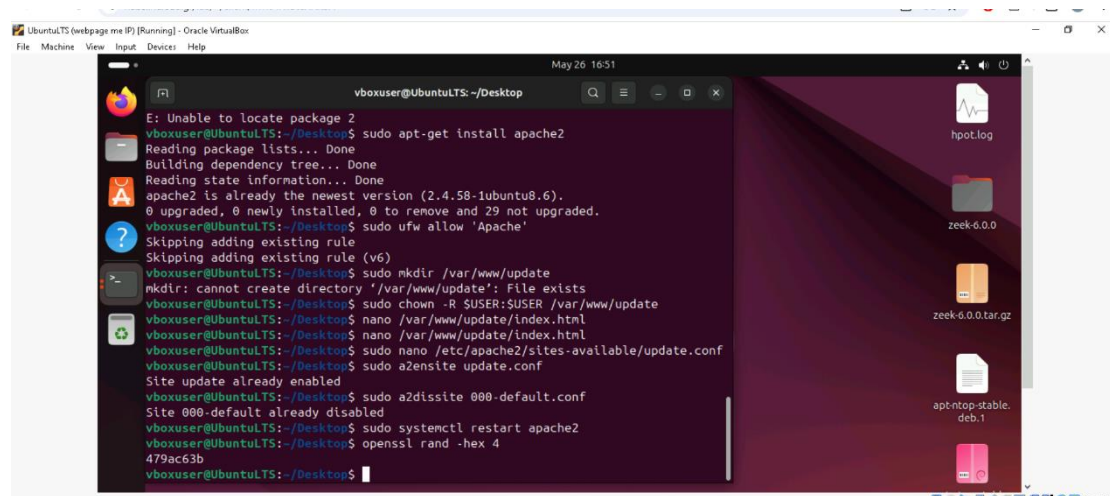


Step 7: Verification of the Setup

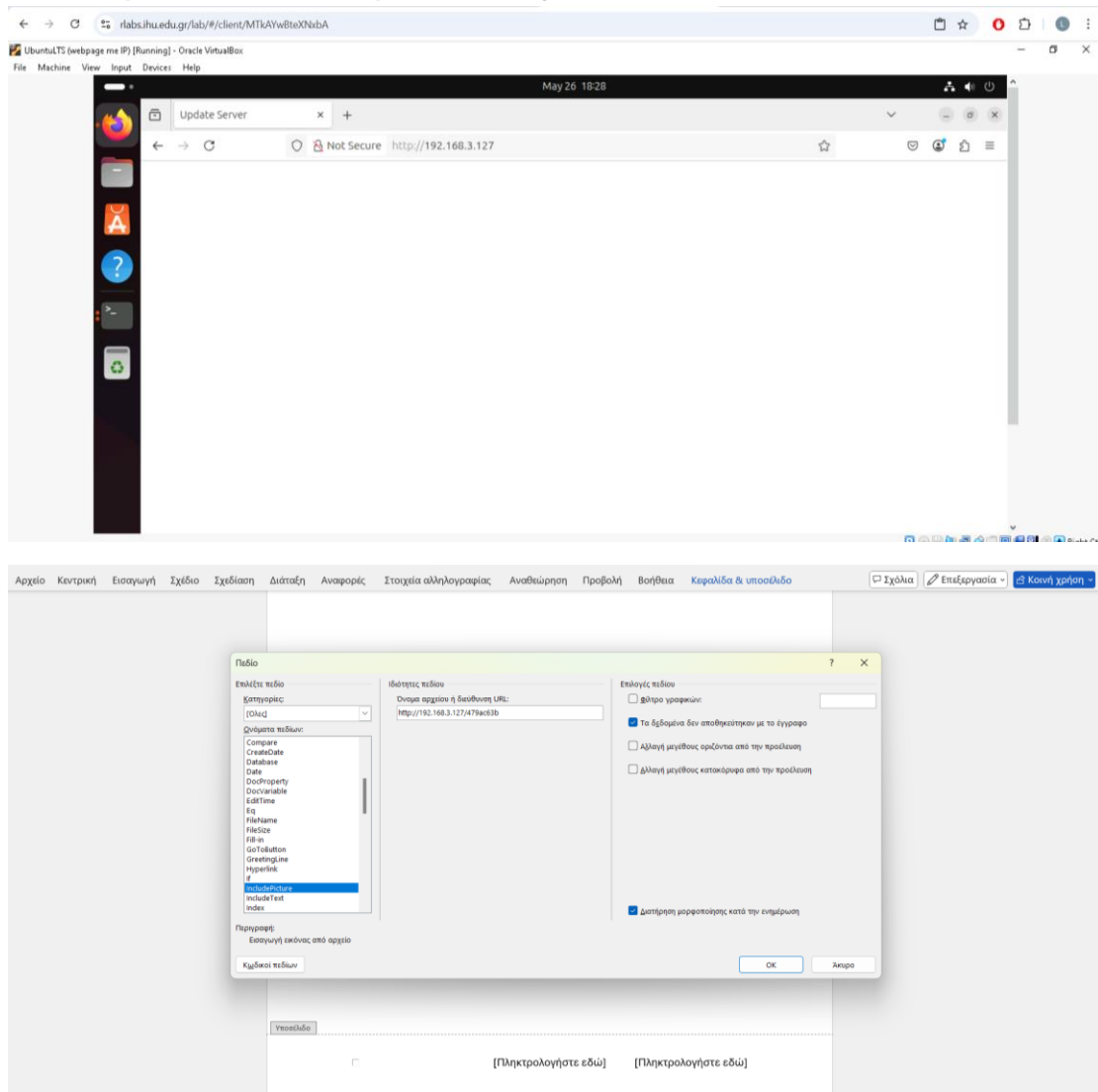


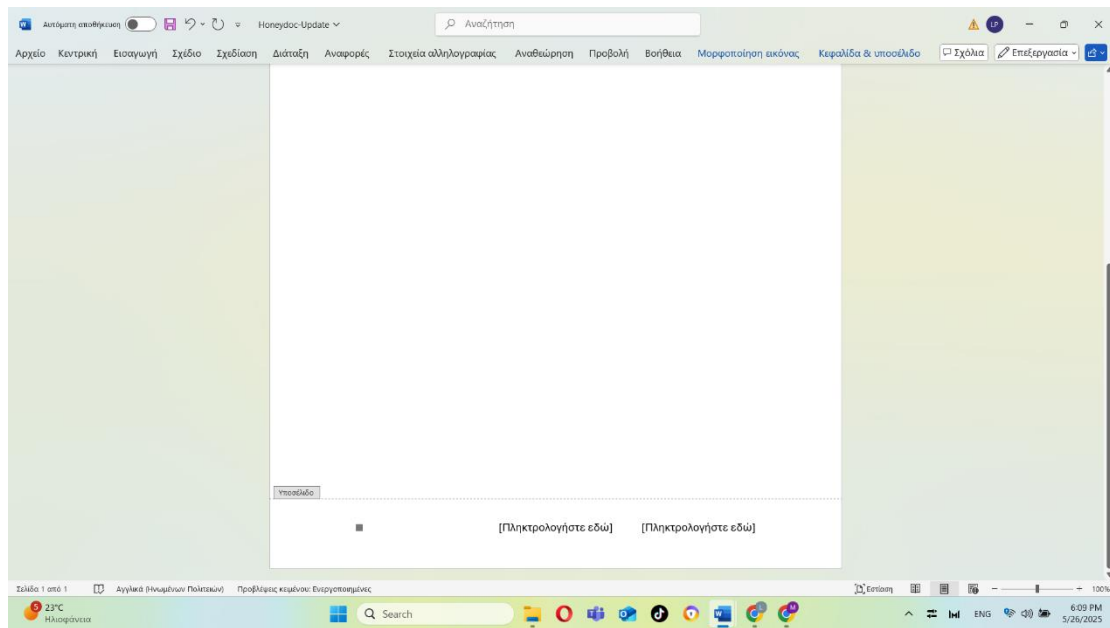
Step8: I generated a Tracking Token URL

openssl rand -hex 4



Step 9: I Created a Word Document and Followed the instructions (I used the IP of my ubuntu machine, because the url didn't work. Also the first screenshots verifies that the ip redirects to the Update Server)





Logging was successful

